Estimating 337 nm photons from RREA electrons

Electron tracks, air density, electric field and the number of photons emitted.

1 Start with a yield per electron-metre

A *relativistic runaway electron avalanche* (RREA) contains energetic electrons that accelerate and multiply in an electric field. A photon count depends not only on their number and spectrum, but also on their paths and low-energy secondaries. Xu reports fluorescence yields per runaway electron per metre. [1, Secs. 2.1 and 4.1]

For approximately uniform conditions and a matching electron cascade,

$$\boxed{N_{\gamma,337} \simeq \eta_{337}(E/n_{\mathrm{air}}, n_{\mathrm{air}})\, \mathcal{L}_R.} \tag{1}$$

η_{337} is the effective yield in **337 nm photons per runaway electron per metre**, including the associated low-energy secondary light.
$\mathcal{L}_R$ is the **total runaway-electron track length**: add the distance traveled by every counted runaway electron inside the chosen region.

This formula reformulates the excitation-and-quenching model. The yield must match the full cascade, composition, temperature and electron counting threshold—not only the two quantities displayed in Eq. (1). [1, Sec. 2.2]

The inputs to write down first

Input	What it means in practice
Electron number N_R	Specify injected, exiting, or instantaneously present electrons; also specify the energy interval used to count them.
Air at altitude z	Obtain molecule density $n_{\mathrm{air}}(z)$, temperature and composition from the chosen atmospheric profile.
Electric field E	Specify its magnitude and spatial extent. Use the reduced field E/n_{air} when comparing densities.
Paths or region size	Specify where electron tracking starts and stops, and whether multiplication continues.
337 nm calibration	Use a band-specific η_{337}, or calculate the excitation events and their probability of producing a 337 nm photon.

If the atmospheric inputs are pressure p and temperature T, the ideal-gas conversion is $n_{\mathrm{air}} = p/(k_{\mathrm{B}}T)$, using pascals and kelvin. Altitude alone is not a complete gas specification.

Why the MeV spectrum is not enough

Most excitation of nitrogen's second positive system, which includes 337 nm, is caused by **low-energy secondary electrons**. In Xu's particular 70 Td RREA calculation, half of the $N_2(C)$ excitation comes from electrons below 12.75 eV. That is a result for that distribution, not a universal threshold. A MeV-only spectrum does not determine this population. [1, Sec. 3.2.1 and Fig. 5b]

ELECTRON ACCOUNTING / FIELD DEPENDENCE

2 Count the paths and match the field

No appreciable multiplication

For N_R electrons that each travel a mean path length ℓ in the selected region,

$$\mathcal{L}_R \simeq N_R \ell, \qquad N_{\gamma,337} \simeq N_R \ell \eta_{337}. \tag{2}$$

For weighted simulation particles, the more general track tally is

$$\mathcal{L}_R = \sum_{a \in \text{runaway segments}} w_a \, \Delta s_a, \tag{3}$$

where w_a is the number of physical electrons represented by the segment and Δs_a is its length in metres. Count every segment once, using the same runaway-energy definition as the yield calibration. Xu's model uses statistical weights to represent large electron populations. [2, Sec. 2.2.3]

When the population avalanches

For a forward-moving avalanche, let $N_R(s)$ count electrons crossing position s. In a uniform region of length L, assume growth by a factor $e \simeq 2.72$ per distance λ:

$$N_R(s) = N_{\text{in}} e^{s/\lambda}, \qquad N_{\text{out}} = N_{\text{in}} e^{L/\lambda}. \tag{4}$$

Integrating this assumed growth law gives

$$\boxed{\mathcal{L}_R \simeq \int_0^L N_R(s)\,\mathrm{d}s = \lambda\big(N_{\text{out}} - N_{\text{in}}\big).} \tag{5}$$

This is a one-dimensional approximation; use actual paths when scattering or escape matters. Exponential multiplication follows Xu, while the track integral is derived here. [2, Sec. 2.2.2]

> **Do not use $N_{\text{out}}L$ as the track length.** Most exit electrons were created partway through the region. An instantaneous electron count also needs a time history or residence time before it can determine integrated emission.

The electric field changes the light per electron-metre

The field affects *both* avalanche multiplication and the low-energy secondary population. In Xu's cases, stronger sustaining fields maintain more electrons around 11–15 eV, increasing nitrogen second-positive emission. Similar high-energy RREA spectra can therefore have different optical yields. [1, Secs. 3.2.1 and 3.3]

Table 1. Xu's broad-window reference yields at ground pressure. These are **300–430 nm totals, not 337 nm yields**. Reduced fields are calculated using $n_0 = 2.688 \times 10^{25}\ \mathrm{m}^{-3}$. [1, Sec. 4.1 and Fig. 5]

E at ground density	E/n_0	Photons / electron / m
4.3 kV/cm	16 Td	6.2
12.5 kV/cm	46.5 Td	10.7
18.8 kV/cm	70 Td	38.3

$1\ \mathrm{Td} = 10^{-21}\ \mathrm{V\,m^2}$. At another density, the same reduced field requires $E = (E/n_{\text{air}})n_{\text{air}}$. The table demonstrates field sensitivity; obtaining 337 nm requires isolating that band from the modeled spectrum, including separation from nitrogen-ion emission. [1, Sec. 2.2]

ALTITUDE / EXCITATION VERSUS QUENCHING

3 Density has two competing effects

More air molecules give more excitation-producing collisions per metre. They also give more *quenching*: collisions that remove the excitation without producing the desired photon. Both effects belong in the estimate. [1, Secs. 2.2 and 4.2]

Probability that an excitation produces a 337 nm photon

In Xu's band-system approximation, write

$$\boxed{p_{337} = g_{337}\frac{A_C}{A_C + Q_C},} \qquad Q_C = k_{C,\mathrm{N_2}} n_{\mathrm{N_2}} + k_{C,\mathrm{O_2}} n_{\mathrm{O_2}}. \tag{6}$$

A_C is the total radiative decay rate of nitrogen in the modeled excited state C. The k values describe quenching collisions. The photon fraction g_{337} selects the 337 nm band from the second positive system; **not every radiative decay belongs to 337 nm**. This fraction must come from a justified spectral model, not be set to one. [1, Sec. 2.2]

For an illustrative 80% N_2/20% O_2 mixture, use Xu's adopted values [1, Sec. 2.2] [2, Sec. 2.2]

$$A_C = 2.0 \times 10^7\,\mathrm{s^{-1}}, \quad k_{C,\mathrm{N_2}} = 1.0 \times 10^{-17}\,\mathrm{m^3\,s^{-1}}, \quad k_{C,\mathrm{O_2}} = 3.0 \times 10^{-16}\,\mathrm{m^3\,s^{-1}}.$$
$$q = n_{\mathrm{air}}/n_0, \quad n_0 = 2.688 \times 10^{25}\,\mathrm{m^{-3}}, \quad b_C \equiv \frac{A_C}{A_C + Q_C} = \frac{1}{1 + 91.392q}.$$

These are illustrative band-system parameters, not an independent state-resolved 337 nm calibration. Temperature and humidity dependence require appropriate extra inputs. [1, Sec. 4.3]

A controlled density comparison

Hold the reduced field, composition, rate coefficients and full electron population per runaway electron fixed. Excitations per electron-metre then scale as n_{air}. Multiplying by the radiative probability gives the *derived* scaling

$$\eta_{337} \propto \frac{n_{\mathrm{air}}}{A_C + k_{\mathrm{eff}} n_{\mathrm{air}}}, \qquad \boxed{\frac{\eta_{337}(q)}{\eta_{337}(1)} = \frac{92.392q}{1 + 91.392q}.} \tag{7}$$

Here $k_{\mathrm{eff}} = 0.8k_{C,\mathrm{N_2}} + 0.2k_{C,\mathrm{O_2}}$. Keep E/n_{air}, not E, fixed for this comparison. Changes in the cascade or secondary escape require recalculation.

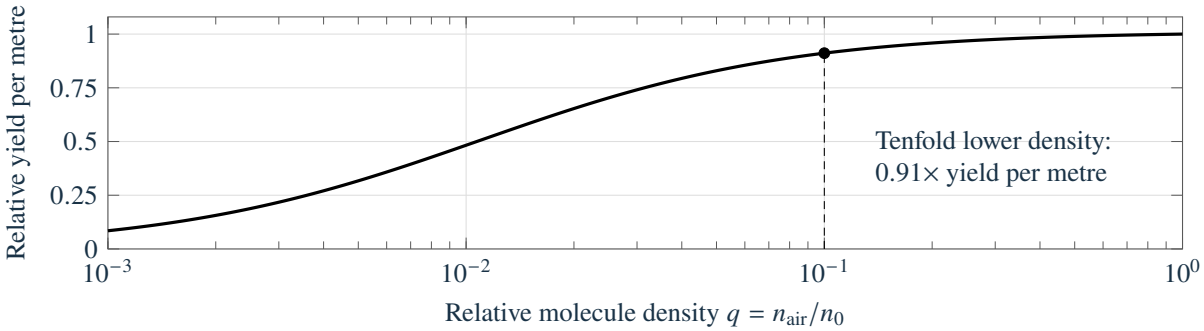


Figure 1. Analytical illustration of Eq. (7), not an altitude profile or new RREA simulation. The yield becomes nearly density-independent when quenching dominates.

Fixed distance versus fixed air column. At $q = 0.1$, the radiative fraction rises from 1.08% to 9.86%, but there are fewer excitations per metre: the yield per metre becomes 0.91 times its reference value. If instead the same electrons traverse the same air column $n_{\mathrm{air}}\ell$, their paths are ten times longer and the total becomes about 9.1 times larger, under the same assumptions.

NUMERICAL IMPLEMENTATION / TWO USABLE ROUTES

4 Obtain the band-specific yield

The missing ingredient in Eq. (1) is the actual 337 nm coefficient for your conditions. Choose one of the following routes. Both account for the secondary-electron light; they are **alternatives, not additive contributions**.

Route A: count excitation events in a full-energy simulation

Follow the relativistic electrons and their low-energy secondaries in the chosen gas and field. Whenever an electron excites $N_2(C)$, record the event's position and particle weight. Xu's Monte Carlo framework can record the location and time of the collisions. [1, Secs. 2.1 and 3.1]

For the direct-excitation, fixed-gas model, the expected total is

$$N_{\gamma,337}^{\mathrm{sim}} \simeq \sum_{a \in \mathrm{N_2}(C)\ \mathrm{excitations}} w_a\, p_{337}(\mathbf{x}_a). \tag{8}$$

Use the local densities in Eq. (6). This tally adds the *expected* photon contribution of every event. Randomly deciding the fate of each excited molecule is unnecessary if only the mean integrated photon count is needed. Any additional feeding channels retained in the model must also be included consistently.

Normalize to what you know. If the run starts with M unweighted seed electrons, including all their descendants in the photon tally,

$$Y_{337}^{\mathrm{seed}} = \frac{N_{\gamma,337}^{\mathrm{sim}}}{M}, \qquad N_{\gamma,337}^{\mathrm{physical}} = N_{\mathrm{in}} Y_{337}^{\mathrm{seed}}. \tag{9}$$

This is a yield **per injected electron for the modeled geometry**. With weighted seeds, divide by the sum of the seed weights instead of M. Simple rescaling assumes unchanged fields and gas conditions.

To obtain an effective yield per electron-metre for uniform, representative conditions, also tally Eq. (3) and form

$$\eta_{337} = \frac{N_{\gamma,337}^{\mathrm{sim}}}{\mathcal{L}_R^{\mathrm{sim}}}. \tag{10}$$

The photon and track tallies must describe consistent regions and electron histories. A coefficient calibrated for one short region, with secondary losses through its boundaries, should not be treated as a universal bulk coefficient.

Route B: use a matched table with high-energy tracks

When your transport code tracks only high-energy electrons, obtain η_{337} from a separate, validated low-energy-cascade calculation. Then accumulate

$$N_{\gamma,337} \simeq \sum_{a \in \mathrm{runaway\ segments}} w_a\, \eta_{337}(E_a/n_a, n_a)\, \Delta s_a. \tag{11}$$

Here n_a is the air-molecule density along segment a. Interpolate only within the calibrated conditions, and keep the runaway-energy cutoff and path-length convention consistent. A local lookup assumes that the secondary cascade resembles the calibration; rapid field changes or boundaries can invalidate that assumption.

A MeV-only excitation tally is not Route B. Applying the nitrogen excitation cross section only to the relativistic electrons misses the dominant secondary contribution. Xu finds second-positive excitation negligible above roughly 1 keV for the relevant cross section. [1, Sec. 3.2.1]

FIELD-FREE COMPARISON / APPLYING THE ESTIMATE

5 Use the right conversion factor

A deposited-energy shortcut outside a sustaining field

For electrons slowing down without a significant accelerating field, a different useful estimate is

$$N_{\gamma,337} \simeq Y_{337}^{\mathrm{dep}}\, E_{\mathrm{deposited}}\,[\mathrm{MeV}], \tag{12}$$

where Y_{337}^{dep} is a photon yield per deposited energy, not per electron.

As a **separate laboratory benchmark**, AIRFLY measured a 337 nm yield of approximately 5.61 photons/MeV in air at 1013 hPa and 293 K, using a proton beam. [3] If a freely decelerating electron population and its secondaries deposit an average of 7 MeV per injected electron in the selected volume at those conditions, the illustrative estimate is

$$N_{\gamma,337} \simeq 5.61 \times 7 \times N_{\mathrm{in}} \simeq 39\, N_{\mathrm{in}}. \tag{13}$$

This uses an *assumed deposited energy*; it does not infer that energy from an RREA spectrum. At another altitude, the fluorescence yield needs the appropriate pressure, temperature and composition correction. Energy carried out of the chosen region is not local energy deposition.

> **Do not use** 39 **photons per electron as an active-RREA coefficient.** In a sustaining field, electrons repeatedly gain energy from the field and the low-energy distribution changes. Xu explicitly distinguishes air-shower fluorescence efficiency from field-driven TGF fluorescence. [1, Sec. 4.1]

Does the 200 microsecond pulse duration change the total?

For a prescribed injection pulse with $\sigma \simeq 50\,\mu$s, changing the injection times while keeping the same electron histories, total injected number, fields and gas conditions changes the light curve, not its integrated photon count. The emitting-state response is fast compared with this envelope. [1, Sec. 2.2] **Do not give every electron a** 200 μs **residence time**: use its actual track inside the emitting region. Evolving fields can change those tracks and the yield.

The calculation to take away

Bottom line: for an active RREA, use **electron track length × a matched 337 nm yield**. Air density and electric field affect the yield and the electron histories. Neither an RREA spectrum nor altitude supplies a universal photons-per-electron number. These are source photons; propagation and detector losses are separate.

References

[1] W. Xu, S. Celestin and V. P. Pasko (2015). *Optical emissions associated with terrestrial gamma ray flashes*. Journal of Geophysical Research: Space Physics, **120**, 1355–1370. doi:10.1002/2014JA020425. Especially Secs. 2.2, 3.2.1, 3.3, 4.1 and 4.2.

[2] W. Xu (2015). *Monte Carlo simulation of terrestrial gamma-ray flashes produced by stepping lightning leaders*. PhD dissertation, The Pennsylvania State University. Especially Secs. 2.2–2.4 and 5.2.1–5.2.2.

[3] M. Ave et al. (AIRFLY Collaboration, 2012 preprint). *Precise measurement of the absolute fluorescence yield of the 337 nm band in atmospheric gases*. arXiv:1210.6734. External laboratory benchmark used only for the field-free comparison.